

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

Question 1 (Healthcare Analytics)

Dataset

Patient	Age	BMI	Blood Pressure	Disease Severity
P1	24	19.8	110	Low
P2	32	24.5	118	Medium
P3	41	28.1	132	High
P4	48	30.3	145	High
P5	56	33.2	152	Critical
P6	65	36.5	168	Critical
P7	38	26.2	124	Medium
P8	29	22.1	116	Low

Task

Construct a visualization in **R (ggplot2)** that:

- Maps **Age** to the x-axis.
- Maps **Blood Pressure** to the y-axis.
- Represents **Disease Severity** using an appropriate color palette.
- Represents **BMI** using point size.
- Justify why your aesthetic mappings and scales effectively communicate health risk.

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization

Question 2 (Climate Science)

Dataset

Month	Solar Radiation	Temperature (°C)	Rainfall (mm)
Jan	180	20	25
Feb	200	23	18
Mar	240	29	12
Apr	280	35	8
May	320	39	5
Jun	300	36	65
Jul	260	32	150
Aug	250	31	170
Sep	240	30	120
Oct	220	28	85
Nov	190	24	45
Dec	170	21	30

Task

Develop **two visualizations**:

- 1. A Cartesian line chart showing temperature variation.
- 2. A Polar coordinate chart representing annual rainfall.

Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization

Question 3 (Business Analytics & Marketing)

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Market Share (%)
A	210	18	24
B	185	14	20
C	260	25	28
D	145	12	15
E	120	10	13

Task

Construct a visualization where:

- Sales are represented by bar height.
- Profit percentage is represented using a sequential color scale.
- Highlight the product with the highest market share using a contrasting color.

Explain why your chosen color palette improves business decision-making.

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization

Question 4 (Transportation Engineering)

Dataset

Direction	Vehicles/hour
North	980
North-East	760
East	1240
South-East	890
South	1420
South-West	1130
West	860

Direction	Vehicles/hour
North-West	720

Task

Using **R Programming**:

- Construct a **Polar Coordinate Chart** representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.
- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

Question 5 (Environmental Sustainability Dashboard)

Dataset

City	AQI	Green Cover (%)	CO ₂ Emission	Population (Lakhs)
A	48	45	1.8	9
B	82	32	3.5	15
C	135	21	5.6	22
D	165	18	7.4	28
E	72	38	2.9	12
F	95	30	4.2	18

Task

Design a visualization dashboard that includes:

- Appropriate aesthetic mappings.
- Suitable scales for continuous variables.
- An effective color palette for AQI categories.
- Justification for every visualization design choice.